Design Theory (Normalization)

First Normal Form (1NF)

A relation is in 1NF if:

- All columns/attributes are single-valued (atomic).
- A primary key exists.
- Functional Dependencies (FDs) are identified.

Table Example (Violation)
In the table below, Phone_Number is multi-valued, which violates 1NF.

Student_ID	Name	Phone_Number
101	Karim	555-1234, 555-5678
102	Rahim	555-9999

Checking Process

1. Look for any cell containing multiple values or "repeating groups."
2. Ensure every row is unique via a primary key.

Conversion Process

To convert to 1NF, expand the table so each cell has only one value. This may involve creating a new row for each multi-valued attribute.

Way 1: Considering Student_ID and Name as Primary Key. Otherwise, we can see Student_ID have duplication.

Student_ID	Name	Phone_Number
101	Karim	555-1234
101	Karim	555-5678
102	Rahim	555-9999

Way 2: Decomposing it into multiple tables. Like –

Student_ID	Name
101	Karim
102	Rahim

Name	Phone_Number
Karim	555-1234
Karim	555-5678
Rahim	555-9999

Second Normal Form (2NF)

A relation is in 2NF if:

- It is already in 1NF.
- No partial dependency exists.
- A partial dependency occurs when: A non-prime attribute (i.e. not part of any candidate key) is functionally dependent on part of a composite primary key, but not on the whole key. ($A \rightarrow C$, $AB \rightarrow D$. Here, AB is the candidate key/primary key. Here, C is a non-prime attribute depends on the A, Which is a part of a candidate key.

How can we find prime, non-prime attribute:

- Prime attribute: part of a candidate key, Non-prime a
-
-
-
- ttribute: Not a part of candidate key.
- Let's say, R=(A,B,C,D) ; F={$A \rightarrow C$, $AB \rightarrow D$}
  Now, Find the attribute closure large combination of determinant attribute,
  (AB)+ = ABCD [Super key, candidate key],
  So, A, B is prime attribute.
  And, Rest C, D is non-prime attribute.

Table Example (Violation)

S_id	Name	Addr	C_id	C_name	Credit	Grade
101	A	DHK	1	ICS	3	A+
101	A	DHK	2	SPL	3	A
102	B	KHL	1	ICS	3	B
102	B	KHL	2	SPL	3	C+

- This Table is not in Second Normal Form.
- For example, the attribute (A, DHK) depends only on S_id = 101, which is part of the composite primary key (S_id, C_id). Since a non-prime attribute depends on only part of the primary key (not the whole key), this is a case of partial dependency, which violates 2NF.

  <u>Then what will be the 2NF:</u>

Primary key: S_id

S_id	Name	Addr
101	A	DHK
102	B	KHL

Primary key: C_id

C_id	C_Name	Credit
1	ICS	3
2	SPL	3

Primary key: S_id, C_id

S_id	C_id	Grade
101	1	A+
101	2	A
102	1	B
102	2	C+

<u>Third Normal Form (3NF)</u>

- Table must be in 2NF.
- Transitive dependency is not allowed.

C_Name	T_id	T_name	Credit
SPL	1	A	3
ICS	2	B	1
OOP	1	A	3
DLD	1	A	3

- In DBMS, a **transitive dependency** happens when a non-key attribute depends on another non-key attribute, which then depends on the primary key, creating an indirect link (e.g., A → B and B → C means A transitively determines C).
- The given table is not in Third Normal Form (3NF) because it contains a transitive dependency, which violates the rules of 3NF.
- In this table, *C_Name (Course Name)* can be considered as the primary key. The attributes T_id (Teacher ID), T_name (Teacher Name), and Credit are all dependent on the course (C_Name).
- However, there is a functional dependency between *T_id and T_name* — meaning that the teacher's name can be determined by their ID.
- This creates a transitive dependency: C_Name → T_id → T_name
- Here, *T_name* is not directly dependent on the primary key (C_Name) but rather indirectly through T_id.
- This violates the 3NF rule, which states that non-prime attributes must depend only on the primary key and not on another non-prime attribute.

<u>3NF of the relation will be:</u>

C_name	T_id	Credit
SPL	1	3
ICS	2	1
OOP	1	3
DLD	1	3

T_id	T_name
1	A
2	B

<u>Practice Problem:</u>

Relation: $R(A, B, C, D, E, F)$
Functional Dependencies (FDs): $AB \rightarrow C, C \rightarrow D, C \rightarrow E, E \rightarrow F, F \rightarrow A$

Check in which normal form - 1NF, 2NF, 3NF, BCNF ?

Solution:

Step 1: Check 1NF (First Normal Form) ***[All attributes must contain atomic (indivisible) values.]***

As there's no multivalued or repeating attributes are mentioned in R, so R is in 1NF.

Step 2: Find Candidate Key(s)

Start with AB: $(AB)^{+} = ABCDEF$; So, AB is a Super key, hence it's a candidate key.

Check if any proper subset of AB is a key:

(A)+ = A; No FD starts with A; So, A is not a key

(B)+ = B; No FD starts with B; So, A is not a key

Only candidate key = AB

Step 3: Identify Prime and Non-Prime Attributes

Prime attributes (part of candidate key): A, B

Non-prime attributes: C, D, E, F

Step 4: Check 2NF (Second Normal Form)

[Relation must be in 1NF, no partial dependency (non-prime attribute should not depend on part of a composite key)]

Check dependencies from part of key:

$A \rightarrow$? Doesn't depend on a part of a composite key.

$B \rightarrow$? Doesn't depend on a part of a composite key.

Only dependency involving key:

- $AB \rightarrow C$ (uses full key); No partial dependency, So, R is in 2NF

Step 5: Check 3NF (Third Normal Form)

[Rule: For every FD $X \rightarrow Y$, at least one must be true:

1. *X is a super key, OR*
2. *Y is a prime attribute]*

Check each FD

FD	Is determinant Super Key?	Is dependent Prime?	Status
AB → C	Yes	No	OK
C → D	No	No	Violation
C → E	No	No	Violation
E → F	No	No	Violation
F → A	No	Yes	OK

Transitive dependency exists: $AB \rightarrow C \rightarrow D, E \rightarrow F$
non-prime attributes depend on non-key attributes, So, it's not in 3NF